MARKET EARTH TRINITY ENGINE — SYSTEM TREASURY & LEDGER AUDIT

ACCOUNT MAPPING & CASH / CAPITAL FLOW MANUAL

Temporal Target: **Personal Day 1 (Moon 2, Day 10)** | Environment: **Local Replica → ICP Mainnet**

SYSTEM BALANCE STATUS	PRIMARY LIQUIDITY ENGINE	AUDITING TOOLING
Dual-Token Reserves	**Continuous Bonding Curve**	**dfx ledger & wallet**
Mecoin + Wecoin + ICP Ledger	CRR = 50% Reserve Anchor	CLI Real-time Query Suite

Treasury Audit Overview

This ledger blueprint maps all developer identities, ICP ledger account IDs, cycles wallets, and smart contract reserve balances across the Market Earth Trinity Engine architecture. Use the embedded CLI commands to query live balances directly from your terminal.

01 System Account Architecture & Ledger Map

The Trinity Engine distributes capital and computational fuel across four primary account layers:

Account Layer	Identifier / Type	Asset Type	Primary Operational Purpose
Developer Identity	`dfx identity get-principal`	ICP / Identity	Root authority for canister deployment & administrative triggers.
ICP Ledger Account	`dfx ledger account-id`	ICP Tokens	Primary cash/utility balance for minting cycles and market liquidity.
Cycles Wallet	`dfx wallet balance`	Cycles (T-Cycles)	Computational fuel powering canister memory, execution, and storage.
Trinity Reserve Canister	`trinity_engine` Canister ID	Mecoin / Wecoin Reserves	Continuous reserve pool anchoring the dual-token bonding curve.

02 Commands to Locate & Query All Cash / Balances

Step 1: Check Local & Mainnet ICP Ledger Accounts

```
# 1. Identify current developer identity and principal ID
dfx identity whoami
dfx identity get-principal

# 2. Get your 64-character hex Ledger Account ID
dfx ledger account-id

# 3. Query your ICP cash balance on the local replica
dfx ledger balance

# 4. Query your ICP cash balance on the ICP Mainnet
dfx ledger --network ic balance
```

Step 2: Check Canister Fuel (Cycles Wallet Balance)

```
# 1. Check local cycles wallet balance
dfx wallet balance

# 2. Check mainnet cycles wallet balance (computational runway)
dfx wallet --network ic balance

# 3. Check specific canister status and cycle reserves
dfx canister status trinity_engine
dfx canister --network ic status trinity_engine
```

03 Bond Reserve Audit Script (`src/scripts/check_balances.sh`)

Run this script during tomorrow's **Personal Day 1** execution to dump a full financial audit of all engine accounts:

```
#!/usr/bin/env bash
echo "========== MARKET EARTH TREASURY AUDIT =========="
echo "Date: $(date)"
echo "------------------------------------------------"
echo "Developer Identity: $(dfx identity whoami)"
echo "Principal ID:       $(dfx identity get-principal)"
echo "Ledger Account ID:  $(dfx ledger account-id)"
echo "------------------------------------------------"
echo "[1] Local ICP Cash Balance:"
dfx ledger balance || echo "Local replica not running."

echo "[2] Mainnet ICP Cash Balance:"
dfx ledger --network ic balance || echo "Mainnet balance query failed."

echo "[3] Local Cycles Wallet Reserve:"
dfx wallet balance || echo "No local cycles wallet."

echo "[4] Trinity Engine Canister State & Reserves:"
dfx canister call trinity_engine getSystemMetrics || echo "Trinity Engine canister not deployed."
echo "================================================"
```

04 Account State Checklist for Personal Day 1

Audit Target	Expected State / Baseline	Action Required
Local ICP Ledger	1,000,000,000.00 ICP (Local NNS Setup)	Verify `dfx nns install --ledger-accounts` initialization.
Cycles Wallet	> 1,000,000,000,000 Cycles	Top up canister cycles if balance drops below threshold.
Trinity Reserve	500,000 Reserve / 1,000,000 Supply	Execute `executeGenesisReset()` to initialize bonding curve.